

Marriage Factors on Cheating

Group 1: Kai Yang | Bilwa Khaparde | Ishaan Singh



Table of contents

01 Introduction

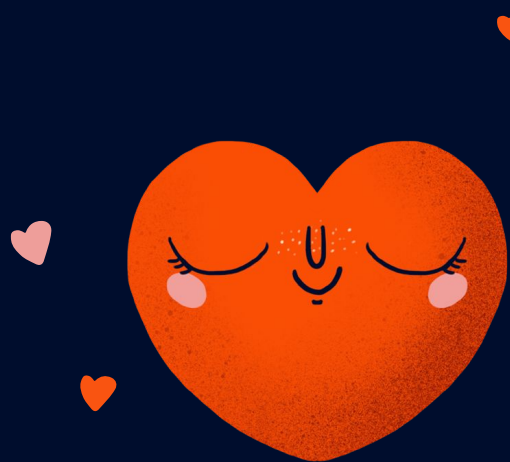
02 Data Description
& Charts

03 Regression
Model

04 Probit
Model

05 Results





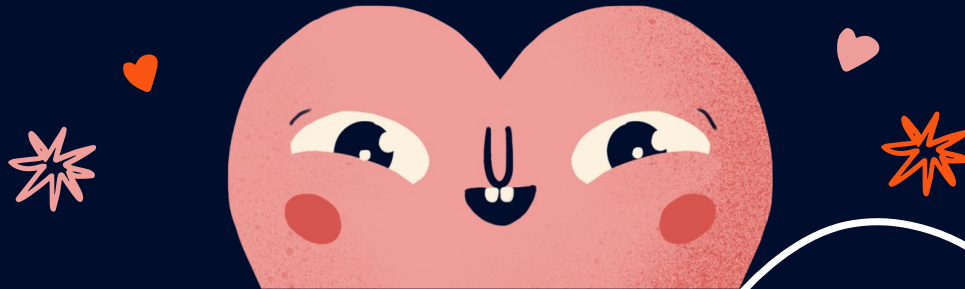
Introduction

What is the effect of rate of happiness, years married and kids on whether a person will have an affair?

Understanding Relationships & Stability

Knowing what factors affect the likelihood of affairs isn't just about predicting infidelity—it's about understanding human relationships, promoting well-being, and addressing societal issues. Whether in psychology, law, counseling, or personal relationships, this knowledge can help create healthier partnerships and stronger families.

-the Internet



Dataset Description

Dataset contains

- 601 observations
- 19 variables

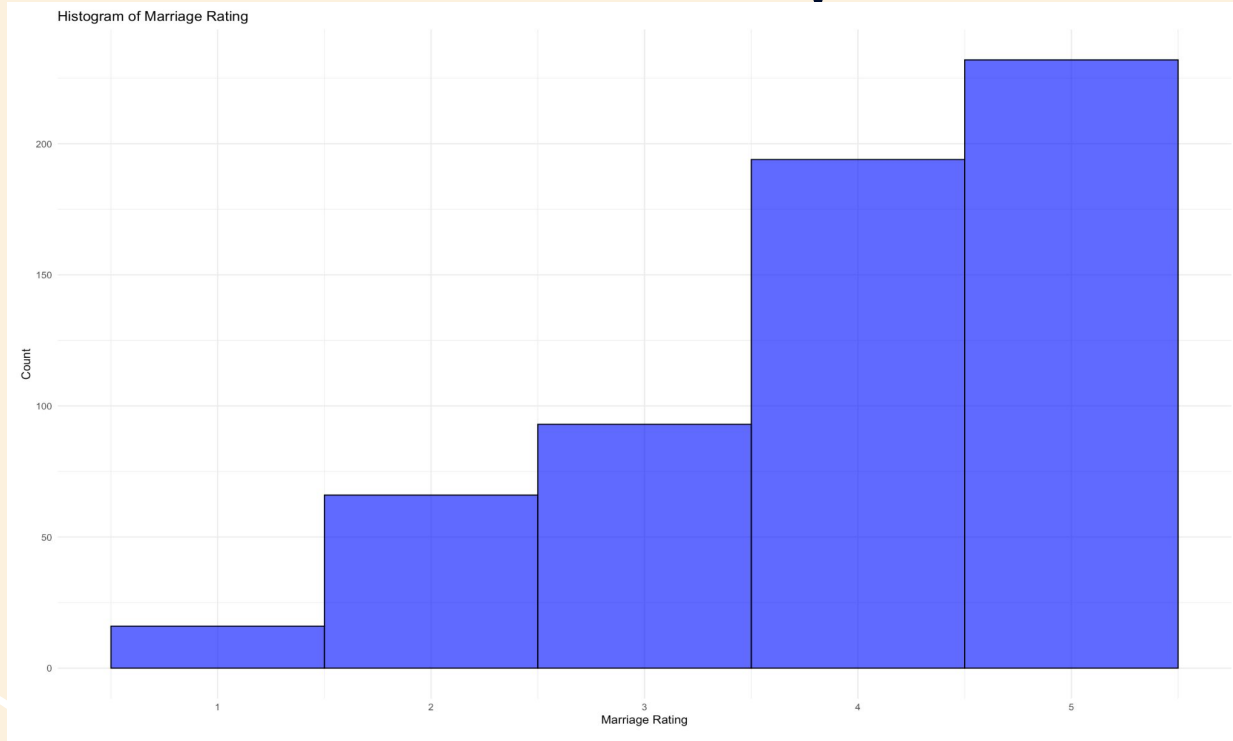
Variables of Interest

Statistic	N	Mean	Median	St. Dev.
Rate of Marriage	601	3.93	4	1.10
Years Married	601	8.18	7.00	5.57
Kids (binary)	601	0.72	1	0.45
naffairs	601	1.46	0	3.30

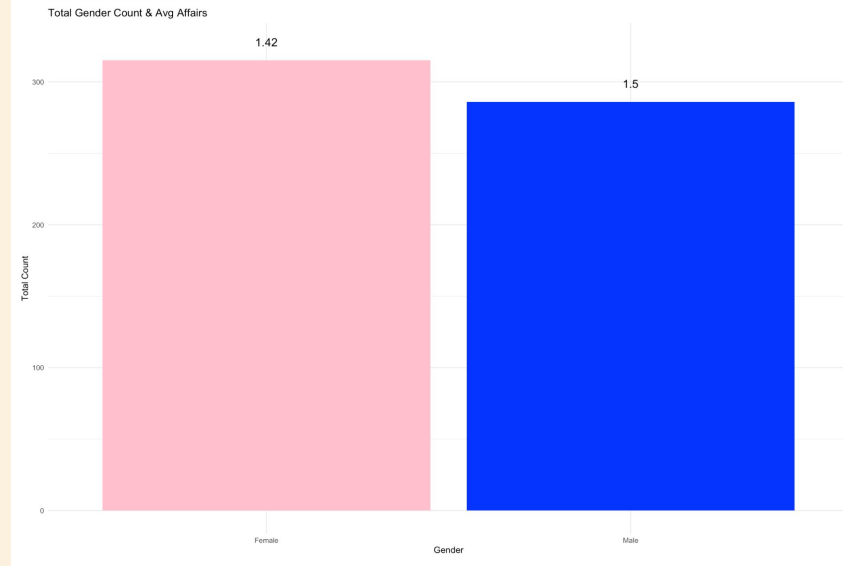
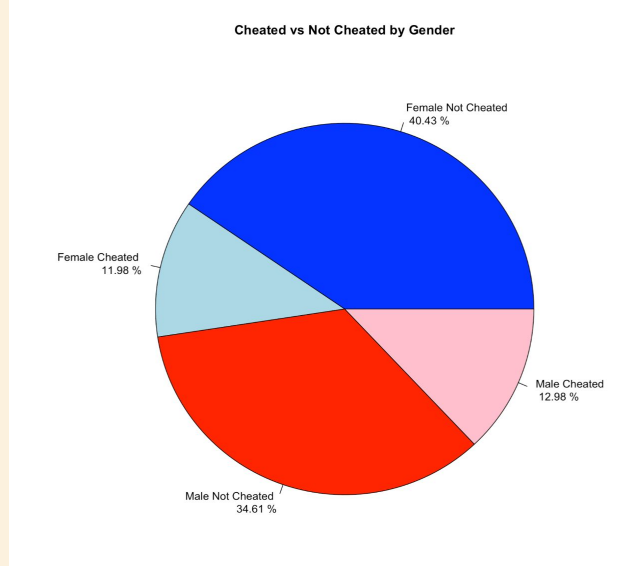
Key variables

- Years married (yrs marr)
- Rate of marriage (ratemarr)
 - On scale of 1-5
- Kids (1 if have kid)
- Age
- Education level (educ)
- Religious (relig)
 - Very, somewhat, slightly, not
- Male (1 if male)

Are You Happy?



gasp CHEATERS?!



Multiple Regressions

Variables of Interest

- Yrsmarr
- Ratemarr
- Kids

Control Variables

- Male
- Age
- Educ
- Religion

Non-linear variables

- Yrsmarr^2
- Educ^2
- Yrsmarr^3
- Educ^3

Added
quadratic terms

Regression Results: What factors influence the number of affairs?

	Dependent variable:								
	naffairs								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
yrs marr	0.111 (0.024)	0.112 (0.033)	0.083 (0.032)	0.083 (0.032)	0.154 (0.044)	0.153 (0.044)	0.171 (0.044)	0.271 (0.129)	0.263 (0.131)
kids		-0.033 (0.379)	-0.188 (0.367)	-0.192 (0.371)	-0.246 (0.366)	-0.234 (0.369)	-0.219 (0.358)	-0.329 (0.423)	-0.323 (0.427)
ratemarr			-0.749 (0.149)	-0.749 (0.150)	-0.755 (0.149)	-0.764 (0.147)	-0.717 (0.147)	-0.715 (0.147)	-0.701 (0.145)
male				0.049 (0.259)	0.216 (0.265)	0.149 (0.277)	0.170 (0.272)	0.164 (0.271)	0.095 (0.284)
age					-0.053 (0.025)	-0.054 (0.025)	-0.049 (0.025)	-0.045 (0.026)	-0.048 (0.027)
educ						0.036 (0.062)	0.018 (0.061)	0.013 (0.062)	-0.708 (0.732)
relig							-0.483 (0.115)	-0.480 (0.115)	-0.492 (0.116)
yrs marr ²								-0.006 (0.008)	-0.006 (0.008)
educ ²									0.023 (0.023)
Constant	0.551 (0.178)	0.562 (0.185)	3.852 (0.685)	3.832 (0.697)	4.956 (0.959)	4.466 (1.386)	5.758 (1.400)	5.542 (1.442)	11.205 (6.187)
Observations	601	601	601	601	601	601	601	601	601
R ²	0.035	0.035	0.094	0.094	0.102	0.102	0.130	0.131	0.133
Adjusted R ²	0.033	0.032	0.089	0.088	0.094	0.093	0.119	0.119	0.120
Residual Std. Error	3.243	3.246	3.148	3.151	3.140	3.141	3.096	3.096	3.095
F Statistic	21.667	10.820	20.548	15.395	13.478	11.282	12.624	11.136	10.085

Note:

NA

Added cubic
terms

Regression Results: What factors influence the number of affairs?

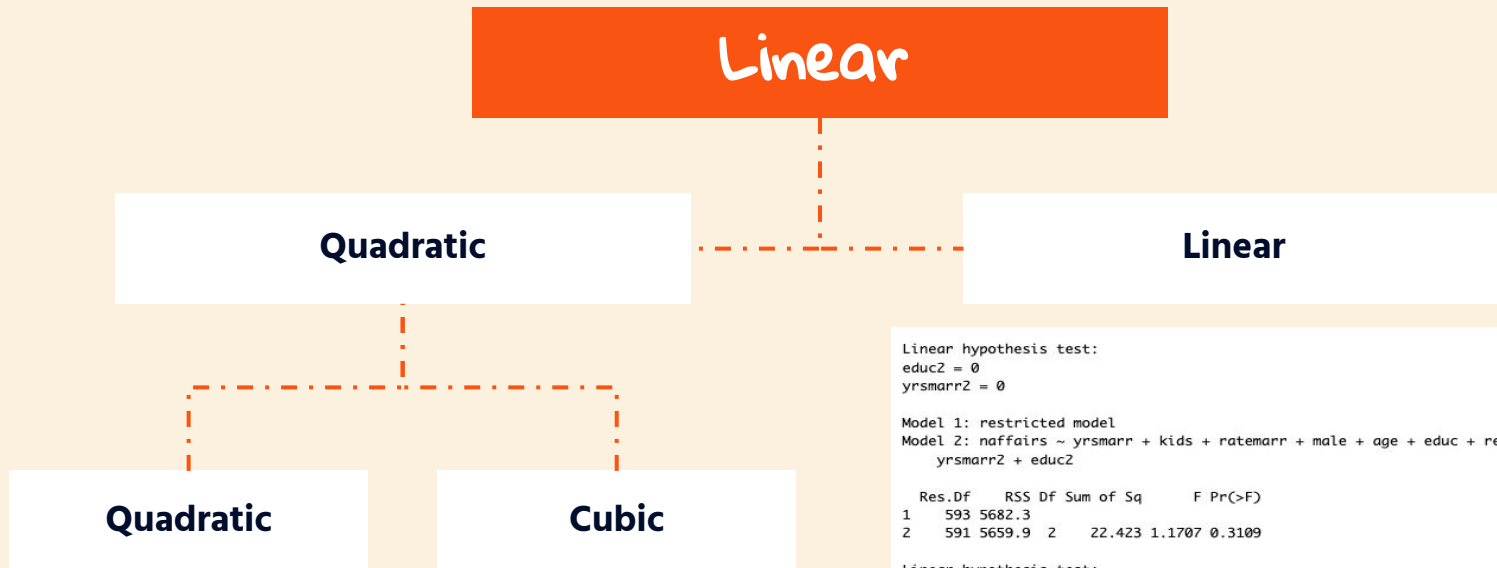
	Dependent variable:									
	naffairs									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
yrs marr	0.111 (0.024)	0.112 (0.033)	0.083 (0.032)	0.083 (0.032)	0.154 (0.044)	0.153 (0.044)	0.171 (0.044)	0.271 (0.129)	0.263 (0.131)	0.143 (0.298)
kids		-0.033 (0.379)	-0.188 (0.367)	-0.192 (0.371)	-0.246 (0.366)	-0.234 (0.369)	-0.219 (0.358)	-0.329 (0.423)	-0.323 (0.427)	-0.284 (0.437)
ratemarr			-0.749 (0.149)	-0.749 (0.150)	-0.755 (0.149)	-0.764 (0.147)	-0.717 (0.147)	-0.715 (0.147)	-0.701 (0.145)	-0.691 (0.145)
male				0.049 (0.259)	0.216 (0.265)	0.149 (0.277)	0.170 (0.272)	0.164 (0.271)	0.095 (0.284)	0.131 (0.286)
age					-0.053 (0.025)	-0.054 (0.025)	-0.049 (0.025)	-0.045 (0.026)	-0.048 (0.027)	-0.051 (0.027)
educ						0.036 (0.062)	0.018 (0.061)	0.013 (0.062)	-0.708 (0.732)	-5.700 (4.404)
relig							-0.483 (0.115)	-0.480 (0.115)	-0.492 (0.116)	-0.486 (0.117)
yrs marr ²								-0.006 (0.008)	-0.006 (0.008)	0.016 (0.045)
educ ²									0.023 (0.023)	0.357 (0.285)
yrs marr ³										-0.001 (0.002)
educ ³										-0.007 (0.006)
Constant	0.551 (0.178)	0.562 (0.185)	3.852 (0.685)	3.832 (0.697)	4.956 (0.959)	4.466 (1.386)	5.758 (1.400)	5.542 (1.442)	11.205 (6.187)	35.591 (22.472)
Observations	601	601	601	601	601	601	601	601	601	601
R ²	0.035	0.035	0.094	0.094	0.102	0.102	0.130	0.131	0.133	0.136
Adjusted R ²	0.033	0.032	0.089	0.088	0.094	0.093	0.119	0.119	0.120	0.120
Residual Std. Error	3.243	3.246	3.148	3.151	3.140	3.141	3.096	3.096	3.095	3.094
F Statistic	21.667	10.820	20.548	15.395	13.478	11.282	12.624	11.136	10.085	8.459

Note:

NA



Which Model Should We Pick?



Linear hypothesis test:
educ2 = 0
yrsmarr2 = 0

Model 1: restricted model

Model 2: naffairs ~ yrsmarr + kids + ratemarr + male + age + educ + relig + yrsmarr2 + educ2

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	593	5682.3				
2	591	5659.9	2	22.423	1.1707	0.3109

Linear hypothesis test:
educ3 = 0
yrsmarr3 = 0

Model 1: restricted model

Model 2: naffairs ~ yrsmarr + kids + ratemarr + male + age + educ + relig + yrsmarr2 + educ2 + yrsmarr3 + educ3

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	591	5659.9				
2	589	5638.4	2	21.49	1.1224	0.3262



Probit Model

Years Married (yrsmarr)

- One additional year of marriage increases the likelihood of an affair by 0.4% to 1.7%.

Kids

- Having children increases the likelihood of an affair by 6.1% to 6.7%.

Rate of Marriage (ratemarr)

- One unit increase in satisfaction decreases the probability of an affair by about 8.2% to 8.4%

Probit n Marginal Effects				
Dependent variable:				
	probit		affair binary model (marginal effect)	
	(1)	(2)	(3)	(4)
yrsmarr	0.0136 (0.0125)	0.0546 (0.0188)	0.0040 (0.0040)	0.0170 (0.0060)
kids	0.2243 (0.1612)	0.2081 (0.1637)	0.0670 (0.0460)	0.0610 (0.0460)
ratemarr	-0.2739 (0.0518)	-0.2727 (0.0534)	-0.0840 (0.0160)	-0.0820 (0.0160)
male		0.1888 (0.1305)		0.0570 (0.0400)
age		-0.0244 (0.0105)		-0.0070 (0.0030)
educ		0.0155 (0.0265)		0.0050 (0.0080)
relig		-0.1861 (0.0519)		-0.0560 (0.0160)
Constant	0.0853 (0.2513)	0.7642 (0.5157)	0.0260 (0.0780)	0.2310 (0.1570)
Observations	601	601	601	601
Log Likelihood	-315.8391	-305.2525		
Note:	NA			

Insights



Insights on Factors Influencing Infidelity



- **Marital Satisfaction and Affairs:** Lower marital happiness is the strongest predictor of infidelity.
- **Children and Infidelity Risk:** Having children slightly increases the likelihood of an affair.
- **Religiosity as a Factor:** More religious individuals are significantly less likely to engage in affairs.
- **Demographic Factors and Infidelity:** Older individuals are less likely to cheat (maturity, shifting priorities).
- **Education and Infidelity:** Education does not significantly predict affairs.



Conclusion

Question

Is there causal relationship between marriage-related factors and cheating?

Answer

Our models show correlation between some marriage-related factors and cheating. However, we are not able to establish causal relationship as there are other confounded variables not added into the model like personality traits and external stressors. The low adjusted R^2 values also indicate the marriage factors we considered in the model only explain a small portion of the variations in cheating behavior.



Thanks!

From **Group 1**

